

VED PATEL

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Education

University of South Florida

Bachelor of Science in Computer Science

May 2028

3.8 GPA

Experience

University of South Florida Information Technology

December 2024 – Present

AI Agent Student Developer

Remote – Tampa, FL

- Architected **AI Agents** using **Copilot Studio** and **Claude 3.5 Sonnet**, automating **2,500+** inquiries via **Graph API** to orchestrate data flows between SharePoint, Teams, and Outlook.
- Developed **custom MCP servers** in **Python** to extend **Claude Code** capabilities, enabling secure **REST API** tool-calling and local file indexing to reduce manual retrieval time by **40%**.
- Engineered **Azure AI Foundry** workflows for **vector-based intent matching**; managed **CI/CD** for **LLM deployment**, reducing misrouting by **85%** via **cosine similarity** and **JSON** logic.

UL Solutions

May 2026 – August 2026

Software Engineering Intern

Fremont, CA

- Developed a **Python based document evaluation pipeline** using **pandas**, **NumPy**, and **scikit-learn** to train a **classification model** that auto evaluated **10,000+** test case outcomes, reducing manual review time by **92%**.
- Architected **automation software** with **subprocess** and **paramiko** to orchestrate multi-device configuration workflows across **Wi-Fi (6E)**, **Bluetooth**, and **LTE/5G** stacks, cutting deployment overhead by **89%**.

UL Solutions

May 2025 – August 2025

Software Testing Intern

Fremont, CA

- Designed and automated **QA test suites** using **pytest** and **Python** with **pyserial** to validate **1,200+ units** across **Wi-Fi (802.11ax)**, **Bluetooth 5.0**, and **LTE/5G** protocols against regulatory compliance standards, eliminating **86% of manual testing** processes.

Projects

Option Pricing Predictor | Python, NumPy, Pandas, Matplotlib, SciPy, Plotly, API

[Live Demo](#) - December 2025

- Developed a **Data Analytics Platform** using **Streamlit**, leveraging **Pandas** for efficient data manipulation and **yfinance API** to pipeline real-time records with a **fault tolerant fallback mechanism**.
- Implemented **SciPy statistical algorithms** to drive **predictive modeling**, utilizing **Matplotlib** to render **static heatmaps** and **NumPy** to optimize vector performance for **heavy computational workloads**.

Autonomous AI Voice Agent | Python, FastAPI, Twilio, Gemini API, JSON

February 2026

- Developed an **autonomous AI caller agent** integrating **Twilio** and **Gemini Multimodal Live API** via **asynchronous Python** and **bidirectional WebSockets**, with a **FastAPI** pipeline enforcing strict **JSON schemas** to route audio payloads for **LLM processing** and structured data extraction from verbal inputs.

Technical Skills

Languages: Python, C++, SQL, JavaScript, HTML, CSS, JSON, PowerShell, KQL

Developer Tools: Azure Portal, Git/GitHub, Power Apps, Copilot Studio, VS Code, Power BI, Jupyter Notebooks, Postman, Azure DevOps

Frameworks & Technologies: Pandas, Pytorch, Numpy, Pytest, Scikit-Learn, Claude Code, Lang Chain CI/CD, LLM, RAG, Azure AI Foundry, Power Automate (Cloud Flows), React, Rest/Fast API, Dataverse

Leadership / Extracurricular

Google Developer Student Clubs

November 2024 – June 2025

Tech Lead

University of South Florida

- Developed and presented Olympic medal prediction model utilizing **linear regression algorithms** to forecast country performance metrics. Engineered **deep learning image classification system** in **Keras**, training **neural networks** on extensive **Kaggle dataset** containing **27,000+ images** to achieve accurate cat-dog classification 95% precision rates.

Association for Computing Machinery

October 2024 – September 2025

Operation Lead

University of South Florida

- Led **32-member executive board**, documenting strategic decisions and coordinating operations.